

Project ID :

25-26J-479

1. Topic (12 words max)

AI-Based Integrated Solution for Smart Pepper Farming in Sri Lanka

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Sri Lanka's pepper industry faces many challenges related to production and selling, including price fairness, knowledge gaps among farmers, no standardized quality assessment and lack of Farmer-to-Buyer Traceability [2].

Lack of Price Uniformity and Fairness- The market lacks standardized pricing mechanisms. Prices differ based on the buyer, often with little consideration for the cost incurred by farmers. This leads to farmer exploitation and revenue instability [1][2].

Limited Knowledge Among New Farmers for Pest and Disease Detection- Many new or small-scale farmers lack access to modern knowledge about pest and disease detection, especially early identification, resulting in lower yields and compromised quality [5][7].

No Standardized Quality Measurement- The absence of specific and standardized criteria to assess pepper quality leads to inconsistent grading and undervaluation of produce [2].

Lack of Transparency- There is minimal transparency from cultivation to the final sale. Buyers often do not have visibility into how the pepper was grown, processed, or transported, reducing trust and traceability in the system [1].

Unpredictable Yields- Most farmers lack early indicators of how much yield a plant will produce. No data driven system exists to forecast yield at the plant level, leading to poor planning of resources, labor.

Problems With Implemented Solutions So Far

- **Low Technology Adoption-** Farmers, especially in rural areas, are not equipped or trained to use complex mobile or digital platforms [7]
- **No End-to-End Traceability-** Existing solutions fail to cover the entire value chain of pepper from plantation, harvesting, and processing, to buyer delivery [6].
- **Generalized Approaches-** Solutions often use a one-size-fits-all model rather than tailoring to the pepper industry's specific requirements, such as quality grading or varietal data [7][8].

This research proposes a smart, transparent, and integrated platform to address the core issues using modern digital technologies. This research aims to address this gap by investigating innovative methods to overcome these problems. In order to Enhance Profitability and Sustainability in Sri Lankan Pepper Industry.

References

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- [2] Institute of Policy Studies, "Raising standards in the Sri Lankan pepper industry," IPS, 2017. [Online]. Available: https://www.ips.lk/wp-content/uploads/2017/03/FT_05May_Raising-standards-in-the-Sri-Lankan-pepper-industry.pdf. [Accessed: 21-Jun-2025].
- [3] Daily Mirror, "Unknown disease spreading across Lankan pepper cultivations," 2016. [Online]. Available: <https://www.dailymirror.lk/business-news/Unknown-disease-spreading-across-Lankan-pepper-cultivations/273-105176>. [Accessed: 21-Jun-2025].
- [4] U. Rathnayaka and R. M. D. B. Rathnayaka, "Blockchain-based traceability system for Sri Lankan agriculture," in Proc. ICACT 2020, University of Kelaniya, Sri Lanka, pp. 56–60. [Online]. Available: <https://fct.kln.ac.lk/media/pdf/proceedings/ICACT-2020/D-8.pdf>. [Accessed: 21-Jun-2025].
- [5] Daily Mirror, "Revolutionizing Sri Lanka's agriculture through AI-driven Agriculture Prompt Engineering," 2024. [Online]. Available: <https://www.dailymirror.lk/amp/business-news/Revolutionising-SLs-agriculture-through-AI-driven-Agriculture-Prompt-Engineering/273-293931>. [Accessed: 21-Jun-2025].
- [6] MIT Solve, "Agriculture Prompt Engineering – Solution Submission," 2024. [Online]. Available: <https://solve.mit.edu/solutions/85194>. [Accessed: 21-Jun-2025].

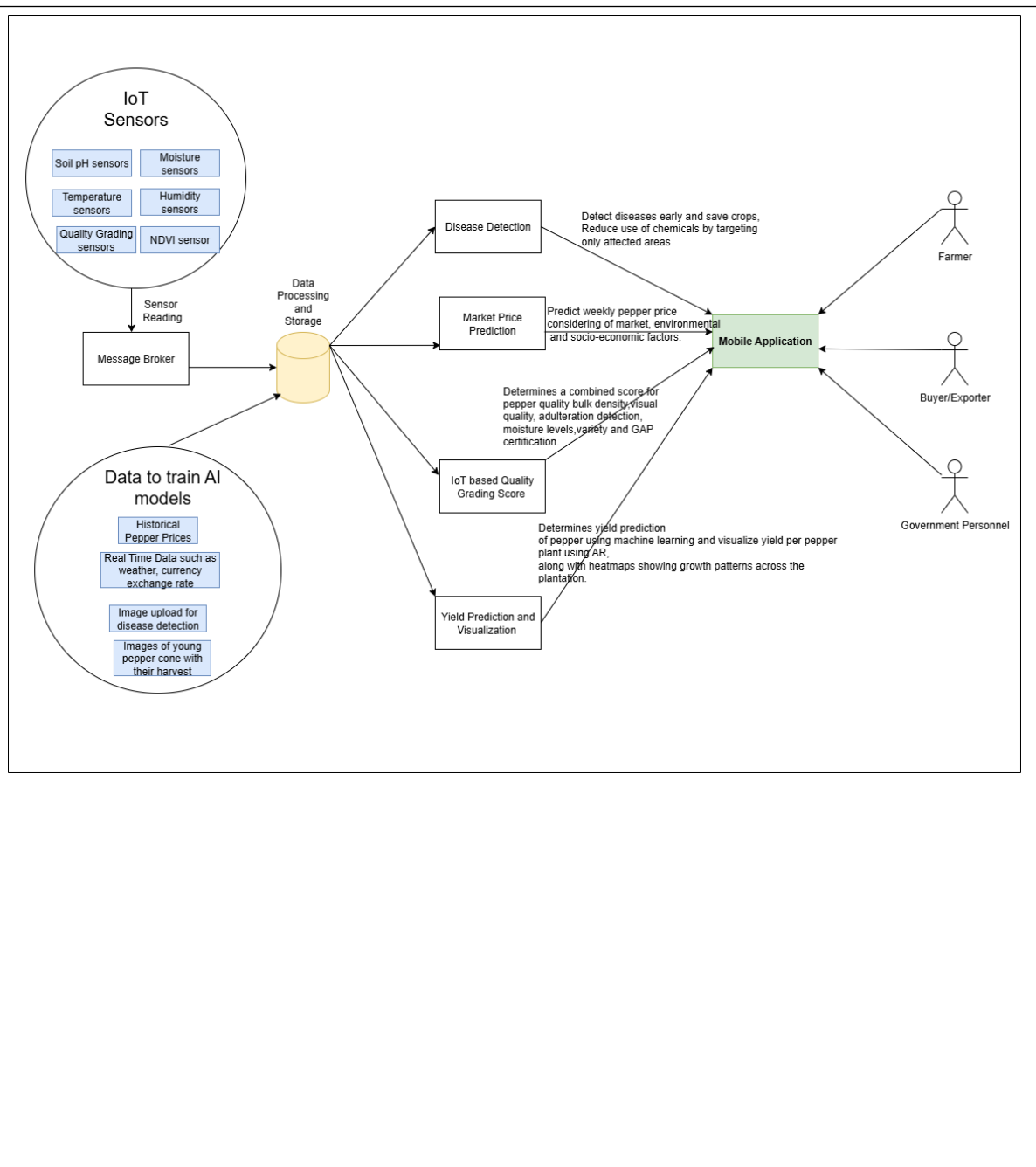
[7] United Nations Development Programme (UNDP), "Blockchain for agri-food traceability," 2021. [Online]. Available: <https://www.undp.org/publications/blockchain-agri-food-traceability>. [Accessed: 21-Jun-2025].

[8] Hector Kobbekaduwa Agrarian Research and Training Institute, "Challenges in the Black Pepper Supply Chain in Sri Lanka," Institutional Report, 2020.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This project proposes a comprehensive AI based integrated solution for smart pepper farming to revolutionize Sri Lanka's pepper industry by enhancing productivity, profitability, and sustainability. The solution integrates four key modules yield prediction and visualization, disease detection, market price forecasting with blockchain-based traceability, and post-harvest quality grading.

1. **Yield Prediction & Visualization:** Using early-stage images of young pepper cones, a CNN-based machine learning model predicts the future yield of each plant. This data is visualized through an **AR-powered mobile app**, allowing farmers to view plant-wise yield directly in the field. Additionally, **spatial heatmaps** are generated to analyze growth distribution across the plantation.
2. **Disease Detection:** Farmers can detect diseases early by uploading leaf images. A hybrid deep learning model uses both RGB images and NDVI values to diagnose plant stress before visible symptoms emerge. The system offers **localized treatment recommendations** and expert support.
3. **Price Forecasting & Traceability:** A **Temporal Fusion Transformer (TFT)** model forecasts weekly pepper prices using multi-dimensional inputs like weather, markets, and exchange rates. Forecasts guide smart selling. All outputs are logged on a **blockchain**, and each pepper batch gets a **QR code** linking to its production and quality history, improving market trust and transparency.
4. **Post-Harvest Quality Grading:** A low-cost IoT device measures physical parameters (bulk density), while image processing checks visual quality. Scores are assigned to classify batches into standard grades, and quality reports are downloadable.



7. Brief description of specialized domain expertise, knowledge, and data requirements
(300 words max)

Domain expertise

This project requires expertise across several specialized areas. Knowledge of agricultural practices and pepper cultivation is essential to understand the real-world challenges faced by farmers, especially in disease management and quality assessment. Skills in Machine Learning, AR based visualization, IoT development, including sensor integration and real time data collection, are vital for monitoring environmental conditions and pepper quality. Expertise in AI and machine learning, particularly computer vision, is necessary for developing models for disease detection, quality grading, and market forecasting. Also, knowledge about blockchain technologies and how to use it. Expertise in AR development is necessary to display digital yield information directly onto real world views of pepper plants. Familiarity with mobile app development and user-friendly interface design ensures accessibility for farmers with limited digital literacy.

Knowledge

The system demands strong knowledge of agricultural supply chain and pepper varietal characteristics. Understanding pest and disease patterns helps support accurate AI-based diagnosis. Knowledge of low-cost hardware sensors supports effective IoT device development. Proficiency in machine learning frameworks and image analysis tools is essential for processing plant images and sensor inputs. Blockchain knowledge to depict the transparency in supply chain. Understanding of rural farmer workflows, digital literacy levels, and mobile usage patterns to build intuitive, multilingual, and possibly offline-capable tools. Knowledge on visualization technologies to use of 3D visualization and heatmaps to represent pepper growth patterns over time and across space.

Data requirements

The system requires a range of datasets for development and testing. These include labeled images of young pepper cones, and the yield gained from respective pepper plants, labeled images of diseased pepper plants for training computer vision models, past record of pepper market prices, images of pepper plants bearing different seed varieties to help the model learn and distinguish between patterns associated with each type. Sensor data on soil, moisture, and pepper sample measurements are needed for grading. Historical and real-time price data, climate records, and socio-economic trends are vital for market prediction. GAP (Good Agricultural Practices) certification records and transaction logs help validate quality and traceability. All datasets must be accurately labeled, diverse, and privacy-compliant to ensure reliability and ethical use.

8. Objectives and Novelty

Main Objective To develop an integrated AI- and IoT-powered intelligent decision support system for smart pepper farming in Sri Lanka, combining yield prediction, disease detection, market price forecasting, post-harvest quality grading, and blockchain-based traceability to enhance productivity, transparency, and profitability for farmers and stakeholders.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Meegama D M IT22097224	To develop a module for yield prediction of pepper using machine learning and visualize yield per pepper plant using AR, along with heatmaps showing growth patterns across the plantation.	- Visit pepper plantations and capture set of images of young pepper cone from a pepper plant. Such that collect from around 200 plants. - Then after harvest is yielded weight of the yield is calculated. - Develop an image-based yield prediction model using machine learning (CNN-based) to predict the yield of each pepper plant from early-stage images of pepper cones. - Integrate an AR-based mobile application to visualize the predicted yield per plant interactively in the field.	- CNN-based approach to accurately predict yield from young pepper cone images. - AR-based mobile interface to visualize predicted yield on each individual pepper plant in an intuitive, spatial manner. - Generating automated, blossom-based heatmaps from aerial or ground-level images to aid field-level growth monitoring and planning.

		• Use images of pepper plantation to generate spatial heatmaps showing the intensity and distribution of pepper growth across the plantation.	
Ranasinghe I C IT22353870	To develop an AI-based system for detecting diseases in pepper plants using computer vision and deep learning, aimed at improving early diagnosis, providing treatment recommendations, and enhancing the health and productivity of pepper crops	• Enable intelligent disease analysis through image uploads, allowing farmers to take a photo of an affected leaf to identify the disease and predict how long it will take to fully spread across the plant. • Provide localized treatment advice, including pesticide names, organic alternatives, dosage instructions, and safety tips - tailored to the specific disease and region. • Allow farmers to contact local agriculture officers or experts directly through the app for advanced support, second opinions, or on-field assistance requests. • Build a digital disease history for each farmer, tracking past infections, treatments used, and	• NDVI (Normalized Difference Vegetation Index) is a value calculated using near-infrared (NIR) and red light reflected from plants to measure their health. Healthy plants reflect more NIR and less red light. • We use NDVI in by a handheld sensor or modified camera to capture NIR and red light from pepper leaves. NDVI is calculated for each scanned plant to detect stress or poor health. • Diseased or stressed pepper plants show lower NDVI values than healthy ones. This helps in detecting early disease signs even before visible symptoms appear. • Benefits for Farmers and Agriculture Detect diseases early and save crops, reduce use of chemicals by targeting only affected

		recovery patterns to support better decision-making in future seasons. • Reward consistent healthy crop reporting with digital badges or certificates, encouraging proper field monitoring and timely reporting. • Enable farmers to share their success stories (with photos or short videos) in the app's community space, helping others learn from real examples. • Offer seasonal reminders and tips, such as how to prevent common diseases during the rainy season or best practices during flowering and harvesting periods. • Smart Notification System Notify users when a specific disease is being reported frequently in their area • Daily Plant Health Tip Show one simple health tip every	areas, monitor plant health in real time, Low-cost, ground-level scanning without drones, Improve pepper crop yield and quality
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		day when the farmer opens the app.	
Ridimahaliyadda A K IT22250674	Predict weekly pepper price using advanced machine learning algorithms, considering a wide range of market, environmental, and socio-economic factors. Integrate blockchain technology to securely store key agricultural insights and links them to a QR code, enabling transparent traceability and trusted verification for each pepper batch in the market	- Gather historical and real-time data on prices, weather, production, exchange rates, and global markets. Historical data are weekly pepper price over multiple years (e.g. 2000–2024) to identify long-term trends and seasonality and Historical exchange rates (USD/LKR) to understand the currency impact on export competitiveness. Real-time Data are Weather forecasts and recent observations and Latest currency exchange rates. - Train and deploy ML models to forecast weekly pepper prices. - Identify seasonality, volatility, and long-term trends to help	- While previous studies have used models such as ARIMA, SARIMAX, Prophet, and even LSTM for agricultural price forecasting, these models often focus on discrete inputs, producing minimal predictions. In contrast, our research introduces the use of Temporal Fusion Transformer (TFT), which is specifically designed for complex, multi-variable time-series forecasting tasks. - It incorporates real-time demand from underlying products to capture consumer-driven price fluctuations a feature often overlooked in traditional forecasting. - TFT's built-in listening and different options ensure that the forecast is interpreted, indicating which variables affect the forecast and

		users understand market behavior. • Provide weekly price forecasts along with reliability indicators to guide user trust. • Recommend smart selling decisions based on forecast trends and threshold logic. • Align and analyze international prices to support export decisions. • Alert farmers/exporters when favorable market conditions are predicted. Blockchain Integration • The system records all outputs from components in yield prediction, disease detection, market forecasting, and grading onto a secure, permissioned blockchain to ensure data integrity. A unique QR code is then generated for	giving users confidence in the results. • Leverage built-in attention and variable selection mechanisms of TFT to provide interpretable predictions, showing which factors most influenced each forecast. • Deploy forecasting results via mobile/web dashboards with clear explanations and “sell or wait” recommendation, helping farmers and exporters make confident decisions. Blockchain Integration • While traditional agricultural platforms may store data centrally, they lack secure, tamper-proof traceability. Our research integrates blockchain to ensure immutable records across all key components—yield, disease, pricing, and grading. • This traceability approach enhances transparency, empowers farmers in
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		each pepper batch, linking to its verified history. When scanned, this QR code provides transparent, tamper-proof information to consumers and exporters, increasing trust and traceability.	pricing and certification, and enables full farm-to-market visibility—features not implemented in Sri Lankan pepper trace systems before. • It integrates yield, disease, price forecasts and grading on blockchain, linking each pepper batch via QR for transparent, immutable traceability.
Adhikari A A J T IT22195548	To develop an AI and IoT-powered module that evaluates the post-harvest quality of pepper based on visual, physical, and certification-based parameters	• Identify and define the critical quality parameters for post-harvest pepper, including bulk density, moisture content, visual appearance, extraneous matter, adulteration detection, variety-based piperine estimation, and GAP certification status. • Design and build a simple, low-cost IoT device using an ESP32 microcontroller, a load cell, and a moisture sensor, that will help farmers accurately measure the bulk density and	• Existing grading systems for pepper often rely on a single factor like bulk density or visual inspection, lacking standardization and transparency. This module introduces a unified, multi-parameter quality scoring system that integrates physical (bulk density, moisture), visual (mold, adulteration), and documentation-based (GAP certification, variety) assessments. This approach is designed to be transparent, farmer-friendly, and aligned with both national (SLS) and international (ESA) standards.

		moisture content of their pepper samples. • Develop image processing models using mobile phone cameras to assess visual characteristics like color uniformity, mold presence, seed adulteration, and overall appearance. • Create a scoring system that assigns points to each factor being evaluated. The combined score will then be used to automatically classify the pepper into a quality grade, such as Grade A, B, or C. • Enable farmers to upload scanned copies of their GAP (Good Agricultural Practices) certification when creating their profile in the app. If verified, this certification will contribute extra points to the overall quality score.	• Manual grading using cups or expensive lab equipment is either inaccurate or inaccessible for most small-scale farmers. This system introduces a novel IoT-based device using ESP32, load cell, and moisture sensor that automates and digitizes bulk density and moisture measurement. It provides consistent, accurate, and traceable results directly linked to the quality scoring framework, enabling widespread adoption at the grassroots level.
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		• Generate downloadable quality reports in PDF format and enable optional QR code generation for traceability integration. • Add a notification system that alerts nearby buyers and exporters whenever a high-quality batch is recorded, helping to create immediate demand and better pricing opportunities for farmers. • Introduce a community leaderboard showcasing the top pepper growers based on quality scores. This gamified approach will motivate farmers to continually improve their practices. • Support farmer learning through clear, step-by-step guidance using visuals, voiceovers, or short videos to explain how each quality factor can be improved over time.	
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Meegama D M IT22097224	The proposed solution for pepper yield prediction and visualization follows a structured Software Development Life Cycle (SDLC). It begins with requirement analysis, data gathering and farmer interviews, especially focusing on challenges faced by pepper farmers, agricultural officers, and buyers. This informs a system architecture that supports image-based machine learning yield prediction and Augmented Reality (AR) visualization. The architecture comprises modules for field data collection via mobile images, machine learning processing (for CNN-based yield prediction) and user interface (mobile app with AR overlay and heatmaps). Each component is designed independently to follow core software engineering principles such as modularity, abstraction, and separation of concerns allowing for future scalability and maintenance. Modularity is achieved by separating the ML model, image pre-processing, visualization, and mobile UI. Abstraction is maintained through REST APIs or microservices that isolate model logic from the user interface. Encapsulation ensures internal image features and model parameters are not exposed to the user but used to generate meaningful insights. Scalability is built-in so the system can support deployment from individual farmers to large plantation datasets. Modeling tools such as UML diagrams are used to visualize the system. Sequence diagrams describe flows such as image upload, prediction, AR visualization. Deployment diagrams illustrate the infrastructure from mobile capture to cloud-based ML processing and AR feedback. The development integrates modern SE practices such as Version control, Automated testing of the prediction model and UI components, Continuous integration/deployment (CI/CD) for iterative improvements, User-centered design, where field pilots and farmer feedback refine the interface (especially AR visual cues, prediction formats, and heatmap readability).
Ranasinghe I C IT22353870	The Pepper Disease Detection Module is a software-driven system developed using the Software Development Life Cycle (SDLC) methodology. It begins with detailed requirement gathering from key stakeholders such as pepper farmers, agricultural officers, plant pathologists, and export quality inspectors. The system focuses on detecting diseases in pepper plants using deep learning models and providing real-time support through a mobile or web application. The system architecture consists of two main layers: the AI processing layer (for image-based disease detection) and the application layer (for user interaction, diagnosis, solutions, and authority

communication). This separation ensures a clean structure that is easy to develop, test, and maintain.

Software engineering principles such as modularity, encapsulation, and abstraction are followed throughout. The deep learning model is trained using a large dataset of healthy and diseased pepper leaf images and operates as an independent module. It performs image classification and returns the disease type and confidence level. This logic is hidden (encapsulated) from the user, while communication is handled through well-defined APIs (abstraction). The mobile app offers a user-friendly interface with offline capabilities, local language support, and accessibility features such as guided image uploads, voice instructions, and visual samples of diseases.

A key feature of the app is the ability to provide treatment recommendations based on the detected disease. For example, if the model identifies a fungal infection, the app suggests the appropriate fungicide, dosage, and application method. These solutions are verified by agricultural experts and regularly updated.

Additionally, users can contact local agricultural authorities directly through the app. This includes options to:

- Submit a report with the image and diagnosis result
- Request field visits or expert advice
- Connect via call, SMS, or WhatsApp
- Track responses and feedback

UML diagrams are used during the design phase:

- Use case diagrams define actions such as “Upload Leaf Image,” “View Disease Report,” “Get Treatment,” and “Contact Authority.”
- Sequence diagrams show how an image is processed, a diagnosis is returned, and how treatment and contact features are triggered.
- Deployment diagrams describe how the mobile app, AI backend, and optional cloud services interact.

From a development perspective, the project uses best practices such as version control (Git), automated testing (for both UI and model accuracy), and continuous integration. It follows an agile methodology, with short development cycles and regular feedback from end users. Security and privacy are ensured through secure image uploads, encrypted user data, and optional anonymous reporting features.

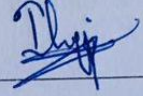
The system aims to achieve:

- High reliability through well-tested AI models

	• High usability through simple interfaces and support for rural users • Maintainability through modular architecture and proper documentation • Real-world impact by empowering farmers with early disease detection, practical solutions, and direct access to expert help
Ridimahaliyadda A K IT22250674	By implementing a thorough and structured methodology to use AI and machine learning to address a domain-specific problem, the Market Forecasting Module is related to the Software Engineering principles. To directly meet the demands of farmers, exporters, and policymakers, it starts with well-defined requirements engineering, outlining goals like pepper price predictions and smart selling recommendations. With the help of a carefully thought-out data pipeline, the module integrates robust data engineering by gathering, cleaning, and integrating a variety of data sources, including local markets, worldwide pricing, and weather APIs. To guarantee dependability, it uses sophisticated model creation and validation methodologies utilizing models in addition to performance indicators and confidence ratings. Separate components for data entry, model training, analysis and alert creation, and the system is constructed modularly to scale and maintain. By means of CI/CD pipelines and API connectivity in the backend to desktop and mobile applications, it is easy to develop and deploy the software. Real-time responsiveness, UI usability, and model stability are guaranteed thorough testing and quality assurance. By protecting user data and guaranteeing accurate projections, the module also places a high priority on security, privacy, and dependability. Furthermore, versioned deployments, user-driven enhancements, and automated retraining processes improve its maintainability. All things considered, this curriculum satisfies existing focus objectives including system thinking, modular architecture, ML integration, and user-centered design while embodying full-stack software engineering methods.
Adhikari A A J T IT22195548	The Pepper Quality Grading Module is designed to follow the principles of the Software Development Life Cycle (SDLC), beginning with detailed requirement gathering from stakeholders such as pepper farmers, exporters, government certification bodies, and buyers. The system is architected to integrate both hardware (IoT devices) and software components (mobile app and AI models), aligning with modular and scalable design principles. The architecture separates the system into three main layers: the hardware layer (ESP32-based device with load cell and moisture sensor), the AI processing layer (for image-based grading and classification), and the mobile

	application layer (for user interaction, offline data handling, and report generation). This separation ensures clear responsibilities and simplifies maintenance. Key software engineering principles such as modularity and encapsulation are maintained throughout the system. The IoT component functions as an independent module that sends standardized data to the mobile app via Bluetooth. The AI models are designed as separate services that handle visual assessment and grading logic. Encapsulation is applied to protect the grading algorithms, and abstraction is achieved through well-defined APIs and data structures. The mobile app is developed following human-centered design practices, offering offline-first capability, support for local languages, and simple interfaces suited for rural farmers. It also includes guided tutorials and visual or audio-based instructions to enhance accessibility. UML tools will be used during the design phase to model system behavior and interactions. Use case diagrams will describe user actions such as “Submit Pepper Batch,” “Upload GAP Certificate,” or “View Quality Report.” Sequence diagrams will illustrate the flow of data from the IoT device through the AI engine to the final grading and report generation. Deployment diagrams will visualize the complete infrastructure, from field-level sensors to the mobile application and cloud storage or notification services. From a development perspective, standard software engineering practices such as version control, automated testing, and continuous integration will be followed. The project will adopt an agile development approach with iterative releases, incorporating user feedback to refine features based on real-world use. Security will be addressed through encrypted Bluetooth communication, secure local data handling, and QR code generation that respects user privacy. The system is designed to ensure reliability (through rigorous testing of both hardware and AI models), usability (by involving end users in the design process), and maintainability (through modular architecture and comprehensive documentation).
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10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms	Ishara	Weerathunga	
Co-Supervisor	Mr	Eishan	Weerasinghe	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes ☒ No ☐

b) Does the proposed topic exhibit novelty?

Yes ☒ No ☐

c) Do you believe they have the capability to successfully execute the proposed project?

Yes ☒ No ☐

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes ☒ No ☐

e) Supervisor's Evaluation and Recommendation for the Research topic:

This topic is suitable for a research project done by undergraduates.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.